Bounded isotropic log-concave samples

Disproof of Conjecture 00000007803

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Abstract

Take independent samples uniform on $[-\sqrt{3}, \sqrt{3}]$. Their common law is centered, isotropic and log-concave. The maximum absolute value is bounded by $\sqrt{3}$ for every sample size, almost surely. Dividing by the square root of the sample size therefore gives zero in the limit, as do the centering and correction terms in the source. The actual normalized laws cannot converge to any nondegenerate Gumbel distribution. The construction uses a genuine infinite independent process and its actual maximum and pushforward laws.

An actual isotropic log-concave law

Let $r = \sqrt{3}$ and let μ be the uniform probability measure on $[-r, r]$. Its density with respect to Lebesgue measure is

$$f(x) = \frac{1}{2r}\mathbf{1}_{[-r,r]}(x).$$

The interval has length $2r > 0$, so this density integrates to one. It is log-concave in the usual extended sense:

$$f(tx + (1-t)y) \geq f(x)^t f(y)^{1-t}, \qquad 0 \leq t \leq 1.$$

For $0 < t < 1$, if x, y lie in the interval, their convex combination does too and the inequality is equality. If either lies outside, the right side is zero. At $t = 0, 1$ the endpoint identity gives the result.

The identity and square functions are integrable on this compact support. Direct integration yields

$$\int x \, d\mu = 0, \qquad \int x^2 \, d\mu = \frac{1}{2r}\int_{-r}^{r} x^2 \, dx = \frac{r^2}{3} = 1.$$

Thus the real one-dimensional random variable is isotropic: it has mean zero and covariance equal to the one-dimensional identity. The source does not exclude fixed dimension or compactly supported laws. Independence below also meets any additional usual independent-sample reading of its sequence hypothesis.

A genuine infinite independent sequence

Use the probability space

$$(\Omega, \mathbb{P}) = (\mathbb{R}^{\mathbb{N}}, \mu^{\otimes \mathbb{N}}), \qquad X_i(\omega) = \omega_i.$$

For every finite index set S and measurable sets B_i, its actual cylinder probabilities are

$$\mathbb{P}\{X_i \in B_i \text{ for all } i \in S\} = \prod_{i \in S} \mu(B_i).$$

Each marginal law is μ, and the coordinates are independent. Consequently every X_i has the density, log-concavity and isotropic moments established above. This is an actual product probability measure, rather than a bounded deterministic sequence used as a proxy.

Each event $\{|X_i| \le r\}$ has probability one. Their countable intersection also has probability one. On that single common event, for every $N \ge 1$ the actual maximum

$$M_N = \max_{0 \le i < N} |X_i| \quad \text{satisfies} \quad 0 \le M_N \le r.$$

In particular $M_N/\sqrt{N} \to 0$ almost surely. This analytic conclusion needs only boundedness, although the actual independent log-concave model above supplies all the conjecture's stated probabilistic hypotheses.

The stated centering and correction also vanish

For sufficiently large N, define the source's terms

$$c_N = \frac{\sqrt{2 \log N}}{\sqrt{N}}, \qquad d_N = \frac{\log \log N}{\sqrt{2 \log N}\sqrt{N}}.$$

Here log is the natural logarithm. Any fixed change of base leaves the vanishing conclusions unchanged. The standard elementary limits

$$\frac{\log N}{N} \longrightarrow 0, \qquad \frac{\log t}{\sqrt{t}} \longrightarrow 0 \quad (t \to \infty)$$

give $c_N \to 0$ and $d_N \to 0$. For the latter, put $t = \log N$ and write $d_N = 2^{-1/2}(\log t/\sqrt{t})N^{-1/2}$. All expressions are ordinary real expressions for $N \ge 3$; arbitrary initial values do not affect a limit.

The normalized law cannot be Gumbel

For any fixed real coefficient a, put

$$Z_N = \frac{M_N}{\sqrt{N}} - c_N - ad_N.$$

This includes the displayed centering without a correction by taking $a = 0$, and either sign of any fixed multiple of the stated correction. On the common probability-one event above,

$$-c_N - ad_N \le Z_N \le \frac{r}{\sqrt{N}} - c_N - ad_N.$$

Both endpoints tend to zero, so $Z_N \to 0$ almost surely. More specifically, the deterministic upper bound is less than one for all sufficiently large N. Therefore the actual pushforward distribution obeys

$$\Pr(Z_N \le 1) = 1 \quad \text{eventually.}$$

A nondegenerate Gumbel law with any location $\ell \in \mathbb{R}$ and scale $s > 0$ has the continuous CDF

$$F_{\ell,s}(x) = \exp\left[-\exp\left(-\frac{x-\ell}{s}\right)\right].$$

In particular $F_{\ell,s}(1) < 1$, since the inner exponential is strictly positive. Convergence in law to this distribution would require convergence of the actual CDF at its continuity point 1, contradicting the eventual identity above.

For completeness, that necessary CDF condition follows directly from the bounded-continuous-function definition of convergence in law. Bound the indicator of $(-\infty, 1]$ above and below by continuous functions that interpolate linearly across intervals of width ε around one. Convergence of their integrals bounds the lower and upper limits of the CDF between $F(1-\varepsilon)$ and $F(1+\varepsilon)$. Continuity at one and $\varepsilon \downarrow 0$ give the required equality.

Theorem 1. *The supplied normalization of maxima of isotropic log-concave sequences need not converge to a nondegenerate Gumbel law, even for independent identically distributed samples.*

Proof. The independent uniform sequence above satisfies the hypotheses. Its actual normalized CDF is eventually one at the finite continuity point 1, whereas every nondegenerate Gumbel CDF is strictly less than one there. This holds with or without the stated correction. □

The first distributional assertion is already false; the additional proposed relation involving L_K, an extreme exponent and a thin-shell constant is not needed. No definition of those extra constants is chosen.

Formalization and reproducibility

The Lean project defines μ as normalized Lebesgue measure restricted to $[-\sqrt{3}, \sqrt{3}]$. It proves the actual density representation, its multiplicative log-concavity inequality, integrability and the exact first and second moments. The infinite sample space uses Mathlib's genuine infinite product measure. Actual finite-cylinder factorization, marginal pushforwards and independence are proved, as are the centered and unit-second-moment coordinate integrals.

The running maximum is defined recursively, and proved equal to the genuine maximum of the first N absolute coordinate values. It is measurable and uniformly bounded on a single probability-one event. Formal little-o logarithm limits prove that both original normalization terms vanish. The resulting almost-sure zero limits hold both without correction and with every fixed real multiple of the displayed correction.

The event CDF is proved equal to Mathlib's CDF of the actual pushforward law. Its value at one is eventually one, and the continuous Gumbel formula is strictly below one. Lean refutes the necessary CDF convergence condition. It does not construct a separate Gumbel measure or formalize the full weak-convergence equivalence; the preceding page proves the required necessity.

The formal index is $n \geq 0$ with sample size $N = n + 1$. The maximum, denominator, center and correction all use that same sample size. This is only a reindexing of the source's first-N-samples statement, with no change to the asymptotic scale.

Lean 4.19.0 and Mathlib revision
`c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From `lean/`, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

Fresh-build logs, source and hashes are included. Printed dependencies use only standard foundational axioms; no auxiliary numerical verifier is required. Compile with `tectonic main.tex`. The author agent and parent reviewed the work internally; no external independent review is claimed.

References

[1] The Last Math Competition, Conjecture 00000007803. Exact bilingual source included as `SOURCE.md`. The counterexample and all model-specific facts are proved above and in Lean.